

MACHINE LEARNING MASTER COURSE

Part 1 Study Guide: ML Foundations & Business Problem Formulation

Target Project: Customer Churn Prediction (Telecom / SaaS Domain)

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Module 1: Introduction to Machine Learning

1.1 What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) focused on building systems that learn patterns from historical data to make predictions on new, unseen data, without being explicitly programmed with hardcoded rules.

1.2 Traditional Programming vs. Machine Learning

Parameter	Traditional Programming	Machine Learning
Inputs	Data + Hardcoded Rules	Data + Historical Answers (Labels)
Outputs	Execution Results / Answers	Learned Model (Rules / Patterns)
Adaptability	Manual code rewrite required for changes	Automatically retrains on new data

Module 2: Business Problem to ML Problem Mapping

2.1 What is Customer Churn?

Customer Churn occurs when a customer cancels their subscription or stops doing business with a company. Acquiring a new customer costs **5 to 7 times more** than retaining an existing one.

💡 Proactive Retention Strategy

Instead of waiting for customers to leave (Reactive), ML predicts who is at high risk of churning *before* they leave, allowing companies to offer targeted discounts (Proactive).

Module 3: Dataset Anatomy (\$X\$ vs \$y\$)

In Supervised Machine Learning, data is divided into two parts:

- **Features (\$X\$):** The input variables/attributes used to make predictions (e.g., Tenure, Monthly Bill, Complaint Count).
- **Target (\$y\$):** The answer or ground-truth outcome we want to predict (e.g., Churn: Yes/No).

Module 4: Data Types in ML

- **Numerical Data:** Numbers used for math (e.g., Monthly Bill: ₹999, Age: 25).
- **Categorical Data:** Words or categories (e.g., Payment Method: UPI/Card, Gender: Male/Female).

Module 5: Key Interview Questions & Answers

Q1: What is the main difference between Features (\$X\$) and Target (\$y\$)?

Features (\$X\$) are the input information given to the model. Target (\$y\$) is the output result we want the model to learn and predict.

Q2: Why must Categorical data be converted into Numerical data?

Computers and ML algorithms rely on mathematical calculations (matrices, linear equations). They cannot perform math directly on text words like "Delhi" or "Male".

Q3: Why is 95% accuracy misleading in an imbalanced dataset?

If 95% of customers stay and 5% churn, a dummy model that predicts "Nobody Churns" gets 95% accuracy but misses 100% of churned customers, causing massive financial losses.